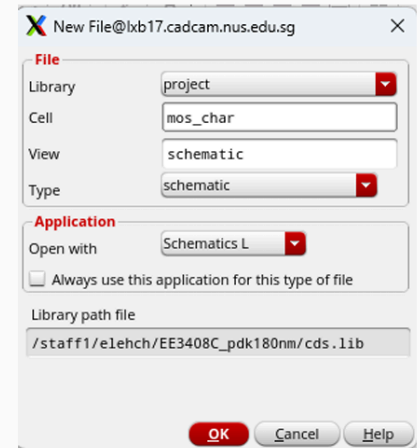
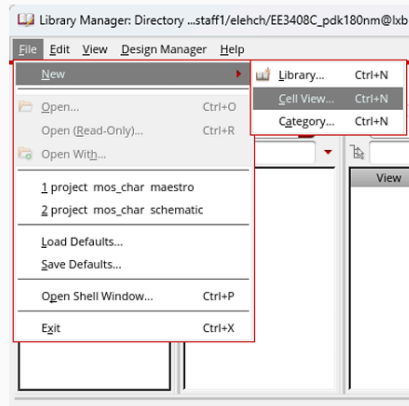


Useful Short-cut Key

- **i**: Insert component
- **c**: Copy component
- **w**: Make wire connection
- **m**: Move the component
- **q**: Edit component properties
- **p**: Create pin
- **l**: Label wire name
- **<shift>E**: Descend edit
- **<ctrl>E**: Return one level up

Create Schematic Cell



- Select “project” library
- Select **File->New->Cell View**
- Select “Schematic” under “Type”
- Type the cell name “mos_char” under “Cell”
- Choose “Schematics L” under “Open with”

- after this step, it gave error with license, and i pressed “session” and ok already

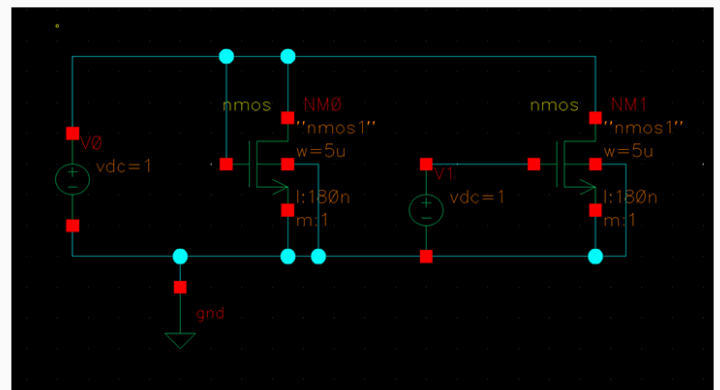
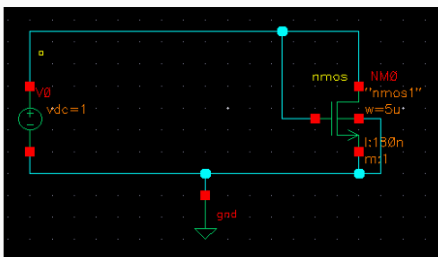
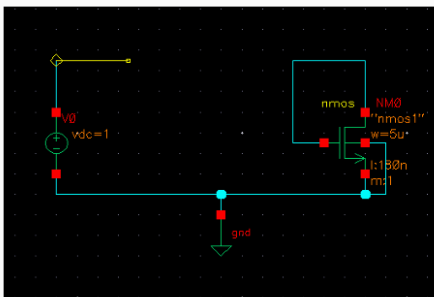
<https://www.quora.com/What-is-vlsi-gpdk-180>

Generic process design kit, 180nm

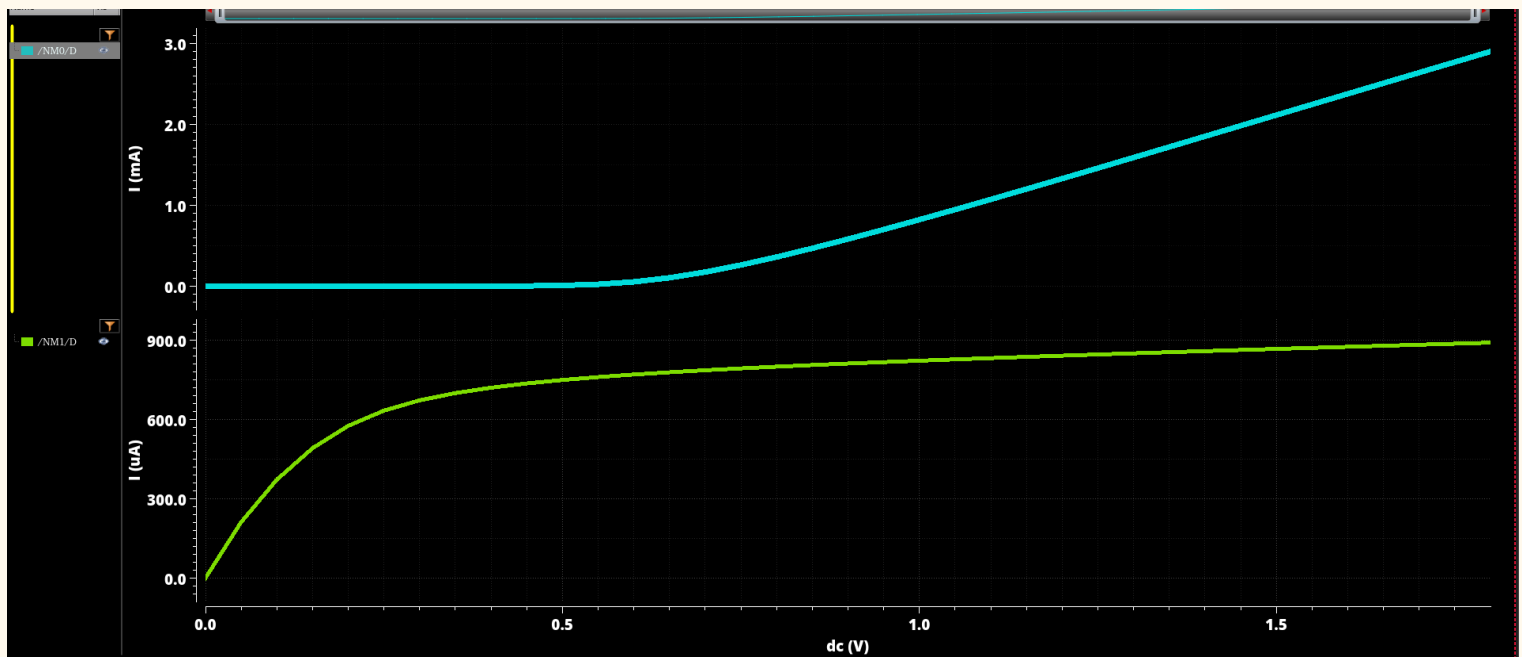
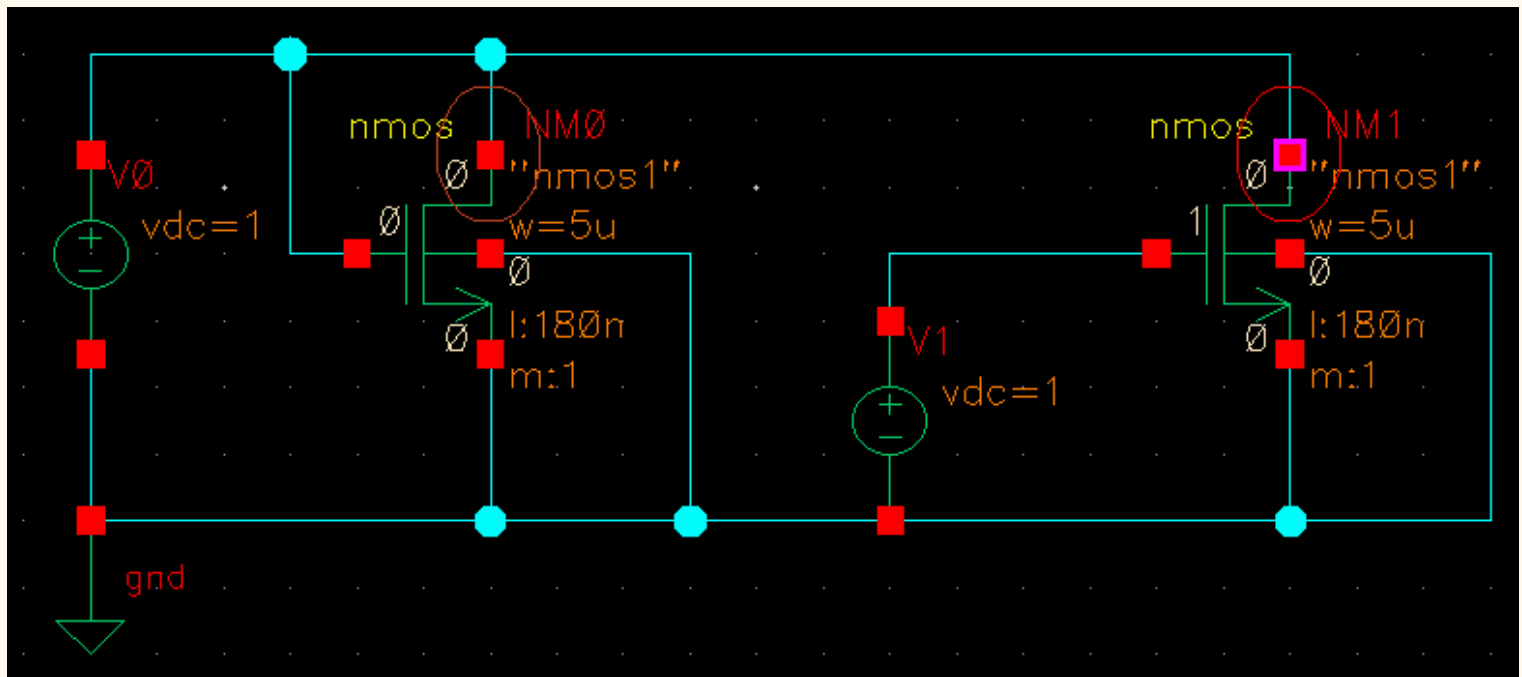
- its used in place of fab house library stuff

What is this lab for

Make Wire Connection



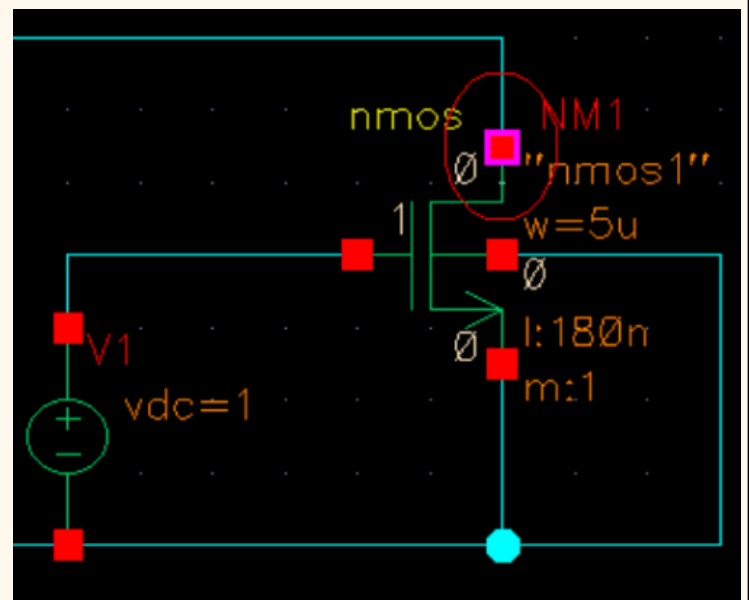
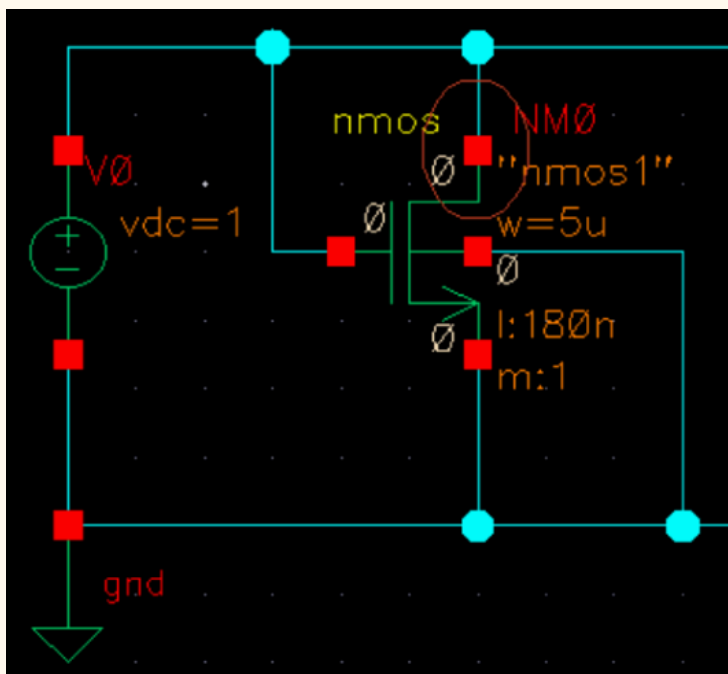
- After placing all components, type “w” to make wire connections between components
- The completed schematic should look like the bottom figure
- Remember to select **File->Check and Save** to ensure no error
- We are trying to extract $\mu_n C_{ox}$, V_{THN} , and λ_n . So try to insert another transistor and vdc source as shown



- the goal is to extract the $\mu_n C_{OX}$, V_{THN} , and λ_n
- This is achieved by measuring the drain current while sweeping the V_{DS}

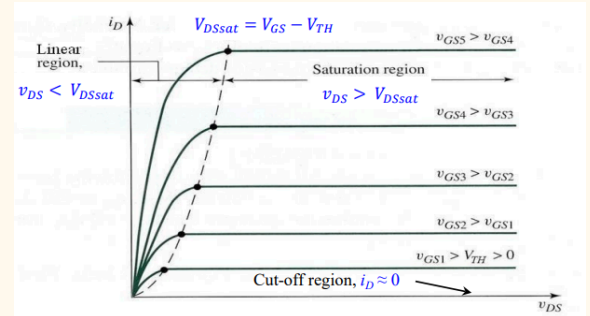
The left mosfet

The right mosfet



- $V_D = V_G$, always in saturation
- this can be used to extract $\mu_n C_{ox}$, V_{THN}
 Saturation Region $v_{GS} \geq V_{TH}$ & $v_{DS} \geq v_{DSsat}$
 $= v_{GS} - V_{TH}$ [$v_{GD} \leq V_{TH}$]
 $i_D = i_{Dsat} = \frac{1}{2} \mu_n \frac{W}{L} C_{ox} (v_{GS} - V_{TH})^2$
 $\sqrt{i_D} = \sqrt{\frac{1}{2} \mu_n \frac{W}{L} C_{ox}} (v_{GS} - V_{TH})$
 $\sqrt{i_D} = \sqrt{\frac{1}{2} \mu_n \frac{W}{L} C_{ox}} v_{GS} - \sqrt{\frac{1}{2} \mu_n \frac{W}{L} C_{ox}} V_{TH}$
- plot $\sqrt{i_D}$ vs $v_{GS} = v_{DS}$, gradient = $\sqrt{\frac{1}{2} \mu_n \frac{W}{L} C_{ox}}$,
 y-intercept = $-\sqrt{\frac{1}{2} \mu_n \frac{W}{L} C_{ox}} V_{TH}$
 - then can solve for $\mu_n C_{ox}$, V_{THN}

- the V_{GS} is fixed at 1V, while the V_{DS} is swept and I_D is measured. This is just the characteristic mosfet plot but only on V_{GS}



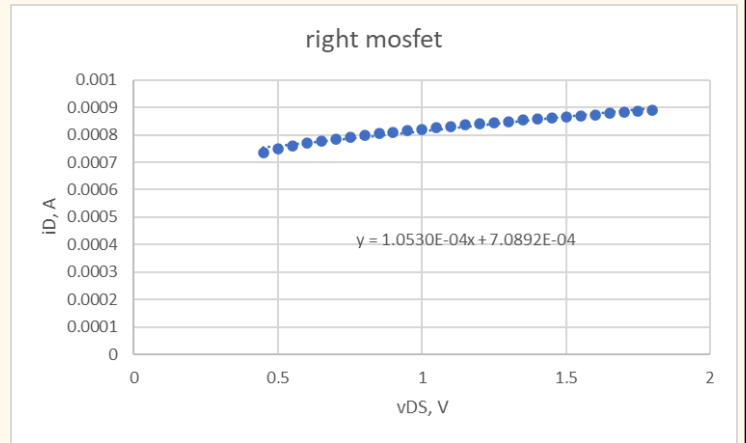
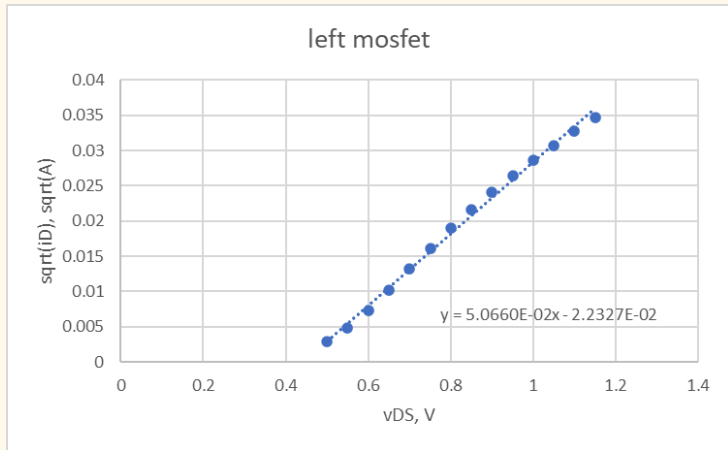
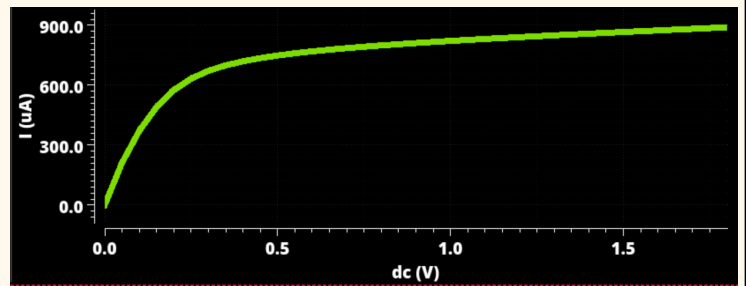
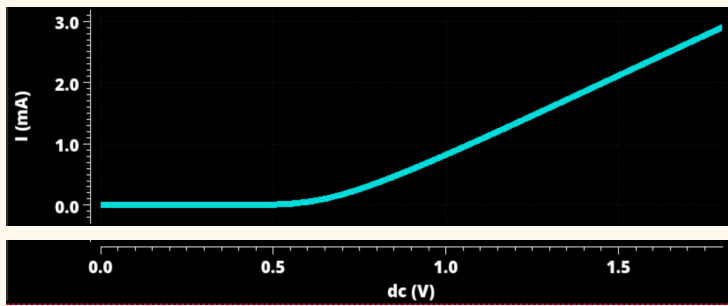
- It covers both the linear and saturation region, the slant on the saturation region will allow the estimation of the λ_n

$$i_D = K_n (v_{GS} - V_{TH})^2 (1 + \lambda v_{DS})$$

$$i_D = K_n (v_{GS} - V_{TH})^2 \lambda v_{DS} + K_n (v_{GS} - V_{TH})^2$$

$$K_n = \frac{1}{2} \mu_n \frac{W}{L} C_{ox}$$

- plot i_D vs v_{DS} when it is in the saturation region
 - then sub in the top formula to solve
 - OR
 - with the output resistance of mosfet amplifier, the gradient of i_D vs v_{DS} at the saturation point is actually the small change in drain current vs small change in $v_{ds} = 1/r_o = \lambda I_D$
 - then the I_D is taken as the current when it starts to saturate



(Following the amount of sample to drop as the lab guide)

$$\sqrt{\frac{1}{2}\mu_n \frac{W}{L} C_{ox}} = 5.0660 * 10^{-2}$$

$$\frac{W}{L} = \frac{5\mu m}{180nm}$$

$$\mu_n C_{ox} = 185 \mu A/V^2$$

$$-\sqrt{\frac{1}{2}\mu_n \frac{W}{L} C_{ox}} V_{TH} = -2.2327 * 10^{-2}$$

$$V_{TH} = 0.44V$$

(Following the amount of sample to drop as the lab guide)

method 1:

$$i_D = K_n (v_{GS} - V_{TH})^2 \lambda v_{DS} + K_n (v_{GS} - V_{TH})^2$$

$$K_n = \frac{1}{2}\mu_n \frac{W}{L} C_{ox}$$

$$\frac{1}{2}\mu_n \frac{W}{L} C_{ox} (v_{GS} - V_{TH})^2 \lambda = 1.0530 * 10^{-4}$$

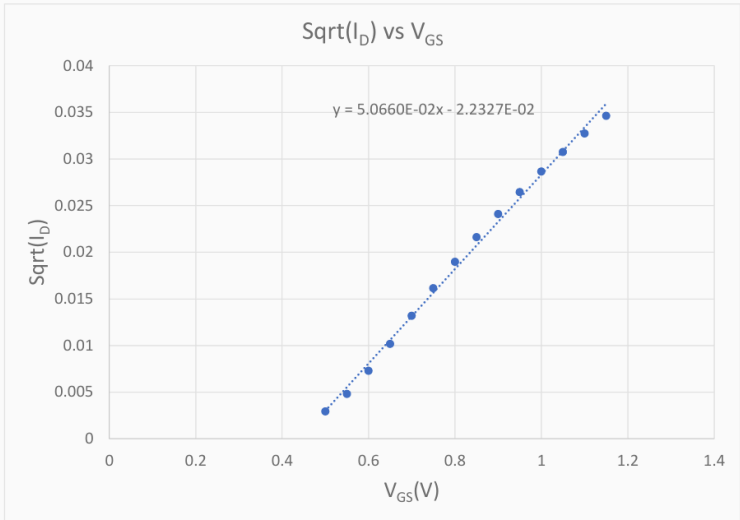
$$\lambda = 0.131$$

method 2:

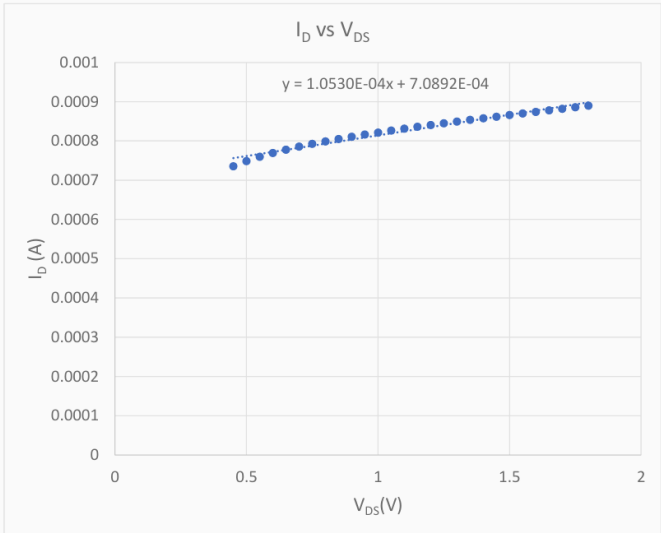
$$\text{gradient} = \lambda I_D$$

taking $I_D =$ when is starts to saturate ($V_{GD} = 0.45$ (with V_{TH} as 0.44), so $V_{DS} =$) =

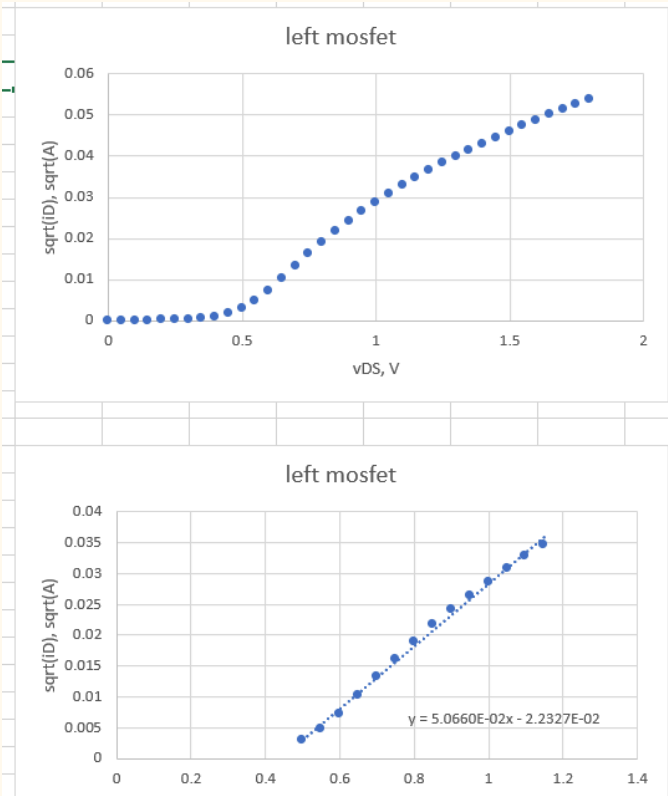
Parameters Extraction



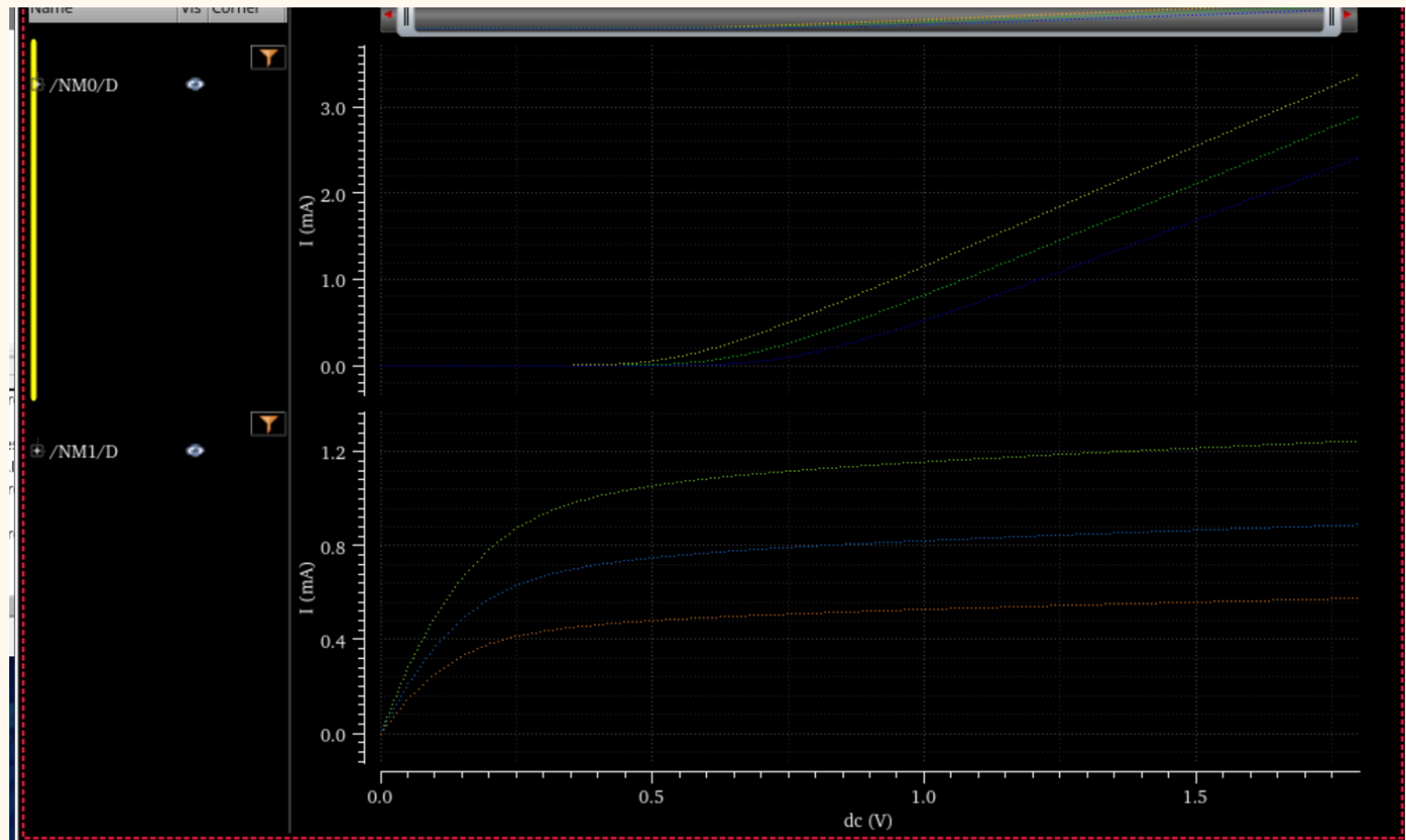
- $\sqrt{I_D} = \sqrt{0.5\mu_n C_{ox} \left(\frac{W}{L}\right)} (V_{GS} - V_{THN})$
- From gradient, we can extract $\mu_n C_{ox}$. From intercept, we can extract V_{THN}
- $\mu_n C_{ox} = 185 \mu A/V$, $V_{THN} = 0.44V$



- $\lambda_n I_D = 1/r_o = \text{Gradient}$
- The current starts to saturate at V_{DS}=0.45V at I_D=735μA
- $\lambda_n = 0.143 \text{ V}^{-1}$



Process Corners



https://en.wikipedia.org/wiki/Process_corners

- its the process variation
- and to study how it would affect circuit performance

FEOL corners [edit]

One naming convention for process corners is to use two-letter designators, where the first letter refers to the N-channel MOSFET (NMOS) corner, and the second letter refers to the P channel (PMOS) corner. In this naming convention, three corners exist: **typical**, **fast** and **slow**. **Fast** and **slow** corners exhibit carrier mobilities that are higher and lower than normal, respectively. For example, a corner designated as **FS** denotes fast NFETs and slow PFETs.

There are therefore five possible corners: typical-typical (TT) (not really a corner of an n vs. p mobility graph, but called a corner, anyway), fast-fast (FF), slow-slow (SS), fast-slow (FS), and slow-fast (SF). The first three corners (TT, FF, SS) are called even corners, because both types of devices are affected evenly, and generally do not adversely affect the logical correctness of the circuit. The resulting devices can function at slower or faster clock frequencies, and are often binned as such. The last two corners (FS, SF) are called "skewed" corners, and are cause for concern. This is because one type of FET will switch much faster than the other, and this form of imbalanced switching can cause one edge of the output to have much less slew than the other edge. Latching devices may then record incorrect values in the logic chain.

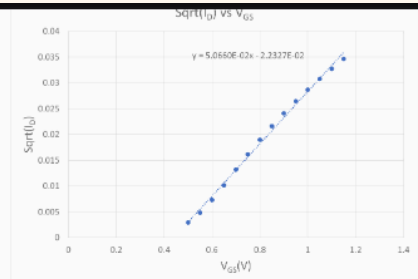
BEOL corners [edit]

Source:[3]

In addition to the FETs themselves, there are more on-chip variation (OCV) effects that manifest themselves at smaller technology nodes. These include process, voltage and temperature (PVT) variation effects on on-chip interconnect, as well as via structures.

Extraction tools often have a **nominal** corner to reflect the nominal cross section of the process target. Then the corners **cbest** and **cworst** were created to model the smallest and largest cross sections that are in the allowed process variation. A simple thought experiment shows that the smallest cross section with the largest vertical spacing will produce the smallest coupling capacitance. CMOS Digital circuits were more sensitive to capacitance than resistance so this variation was initially acceptable. As processes evolved and resistance of wiring became more critical, the additional **rcbest** and **rcworst** were created to model the minimum and maximum cross sectional areas for resistance. But the one change is that cross sectional resistance is not dependent on oxide thickness (vertical spacing between wires) so for **rcbest** the largest is used and for **rcworst** the smallest is used.

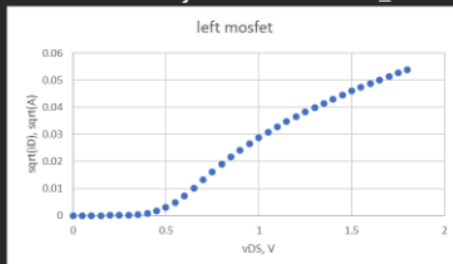
All Q and no A



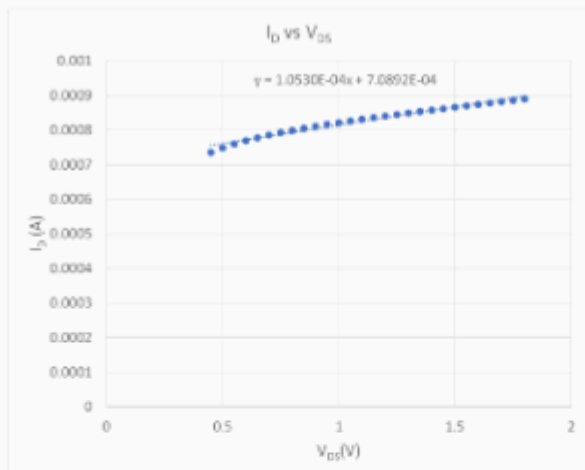
- $\sqrt{I_D} = \sqrt{0.5\mu_n C_{ox} \left(\frac{W}{L}\right)} (V_{GS} - V_{THN})$
- From gradient, we can extract $\mu_n C_{ox}$. From intercept, we can extract V_{THN}
- $\mu_n C_{ox} = 185 \mu A/V$, $V_{THN} = 0.44V$

1) from page 22

- For the parameter extraction of $\mu_n C_{ox}$, V_{THN} , only the points between $V_{GS} = 0.5$ to 1.15 V is taken for the linear best fit, although the entire sweep is from $V_{GS} = 0$ to 1.8 V
- It is clear that excluding $V_{GS} < 0.5$ V is to get rid of the points near cutoff region ($V_{THN} = 0.44$ V)
- If I understand correctly the exclusion of $V_{GS} > 1.15$ V is due to the curve not following the square law



- plotted from simulated data
 - after turn on, this ought to look quadratic going up, however it starts to taper downwards
 - hence, when the square root is taken, the part that is linear is just $V_{GS} = 0.5$ to 1.15 V
- d. My question is: what is causing this effect where the square law is wrong?
- it diverge from square law with just a few volts of v_{DS} and few mA of i_D , which seems very unintuitive
 - it should not be λ since that should make it curve upwards more
 - I don't think there is any error in my simulated data, as I was able to replicate the same exact numerical results as shown in the guide



- $\lambda_n I_D = 1/r_o = \text{Gradient}$
- The current starts to saturate at $V_{DS}=0.45\text{V}$ at $I_D=735\mu\text{A}$
- $\lambda_n=0.143\text{ V}^{-1}$

from page 22

2)

- $V_{TH} = 0.44\text{V}$
- to saturate: $v_{GS} \geq V_{TH}$ & $v_{DS} \geq v_{DSsat} = v_{GS} - V_{TH}$
 $v_{DS} \geq v_{GS} - V_{TH}$
 $v_{DS} \geq 1 - 0.44$
 $v_{DS} \geq 0.56$
- why was $v_{DS} = 0.45\text{ V}$ chosen instead?